

Voice Agent Performance Optimization Roadmap

From Production-Credible → Industry-Leading Voice Agent

Current State

Stack

- Deepgram Nova-3 STT
- Groq Llama 3.1 8B Instant
- ElevenLabs Flash v2.5
- Silero VAD
- Barge-In
- Per-Turn V2V Logging

Current Performance

- P50: 1091ms
- P95: 1439ms

Target Performance

- P50: <800ms
- P95: <1200ms
- Barge-In: <250ms

Gap

- P50 Improvement Needed: ~291ms (26.7%)

Objective

Create a premium enterprise-grade voice agent that feels instantaneous, interruptible, and conversational while maintaining production reliability and scalability.

Phase 1 — Observability & Measurement

Goal

Understand exactly where latency exists before optimizing.

Deliverables

Per-turn latency instrumentation:

```
User Speech Start
↓
VAD End
↓
STT First Token
↓
STT Final Transcript
↓
LLM First Token
↓
LLM Complete
↓
TTS First Audio
↓
TTS Complete
↓
Playback Start
```

Metrics

Track:

- STT Latency
- LLM Latency
- TTS Latency
- Playback Latency
- End-of-Speech Detection
- Barge-In Delay

Success Criteria

Every voice turn contains a complete timing profile.

Phase 2 — VAD & Endpointing Optimization

Goal

Reduce user wait time after speaking.

Problem

Most voice agents wait too long before deciding the user is finished talking.

Actions

Optimize:

- Silence threshold
- Endpoint detection
- Speech confidence thresholds
- Barge-in detection

Target

Current:
400-700ms endpoint delay

Target:
200-350ms endpoint delay

Expected Gain

100-300ms

Phase 3 — Streaming Pipeline

Goal

Remove sequential processing bottlenecks.

Current Model

```
User
↓
STT Complete
↓
LLM Complete
↓
TTS Complete
↓
Playback
```

Future Model

```
User
↓
Partial STT
↓
Streaming LLM
↓
Streaming TTS
↓
Playback
```

Benefits

- Faster perceived response
- Reduced dead air
- More natural conversation

Expected Gain

100-250ms

Phase 4 — TTS First Audio Optimization

Goal

Speak sooner.

Current Problem

Users wait for full text generation before hearing anything.

Improvements

Use:

- ElevenLabs Streaming API
- First sentence streaming
- Incremental synthesis
- Response chunking

Example

Bad:

```
(wait 1 second)
"Sure, I can help with that."
```

Good:

```
"Sure..."
(rest streams in)
```

Expected Gain

100-200ms

Phase 5 — Barge-In Hardening

Goal

Allow users to interrupt naturally.

Requirements

While TTS is playing:

- Continue listening
- Detect interruptions
- Stop playback immediately
- Cancel stale responses

KPI

```
Barge-In Response
<250ms
```

Benefits

- More human conversations
 - Less frustration
 - Better perceived intelligence
-

Phase 6 — Fast Intent Router

Goal

Avoid sending every request through the primary LLM.

Architecture

```
graph TD
    A[Voice Input] --> B[Intent Router]
    B --> C{Simple Intent?}
    C -- Yes --> D[Execute]
    C -- No --> E[LLM]
```

Examples

Simple:

- What's the time?
- Open dashboard
- Check Rainmaker

Complex:

- Analyze market conditions
- Explain a strategy
- Summarize a meeting

Benefits

- Reduced LLM load
- Faster responses
- Lower cost

Phase 7 — Parallel Processing

Goal

Remove unnecessary waiting.

Current

```
Memory
↓
Profile
↓
Tools
↓
Response
```

Future

```
Memory
  \
Profile — Parallel
  /
Tools
```

Parallel Tasks

- Memory Retrieval
- Profile Lookup
- Tool Discovery
- Permission Validation
- Context Retrieval

Benefits

- Lower latency
- Higher throughput

Phase 8 — Tool Latency Management

Goal

Prevent slow tools from making the agent feel broken.

Strategy

Immediate acknowledgment:

"Checking that now."

while tools execute.

Features

- Tool timeouts
- Retries
- Fallbacks
- Caching
- Progress updates

KPI

Tools never block conversational flow.

Phase 9 — Agent Intelligence Layer

Goal

Move beyond speed into quality.

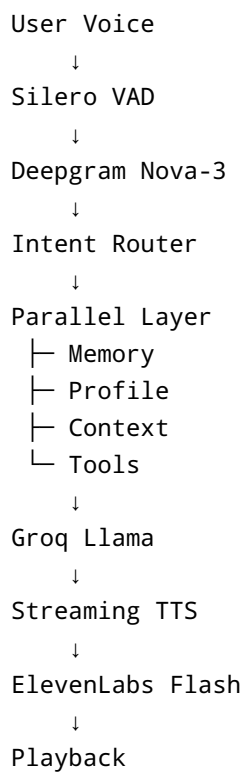
Evaluate

- Intent accuracy
- Tool selection
- Response quality
- Hallucination rate
- Interruption handling
- Context retention

Metrics

Intent Accuracy
Tool Success Rate
Conversation Completion Rate
User Satisfaction
Average Turn Count

Future Industry-Leading Architecture



Success Definition

Minimum Enterprise Grade

- P50 < 1000ms
- P95 < 1500ms
- Barge-In < 400ms

Premium Enterprise

- P50 < 800ms
- P95 < 1200ms
- Barge-In < 250ms

Industry Leading

- P50 < 600ms
- P95 < 1000ms

- Barge-In < 150ms
 - Full streaming pipeline
 - Parallel tool execution
 - Context-aware routing
 - Real-time interruption recovery
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Recommended Execution Order

1. Instrumentation
2. VAD Optimization
3. Streaming STT → LLM → TTS
4. TTS First Audio
5. Barge-In Hardening
6. Intent Router
7. Parallel Processing
8. Tool Latency Controls
9. Agent Intelligence Evaluation

Focus first on latency. Once the system feels instantaneous, optimize intelligence, memory, and reasoning quality.